

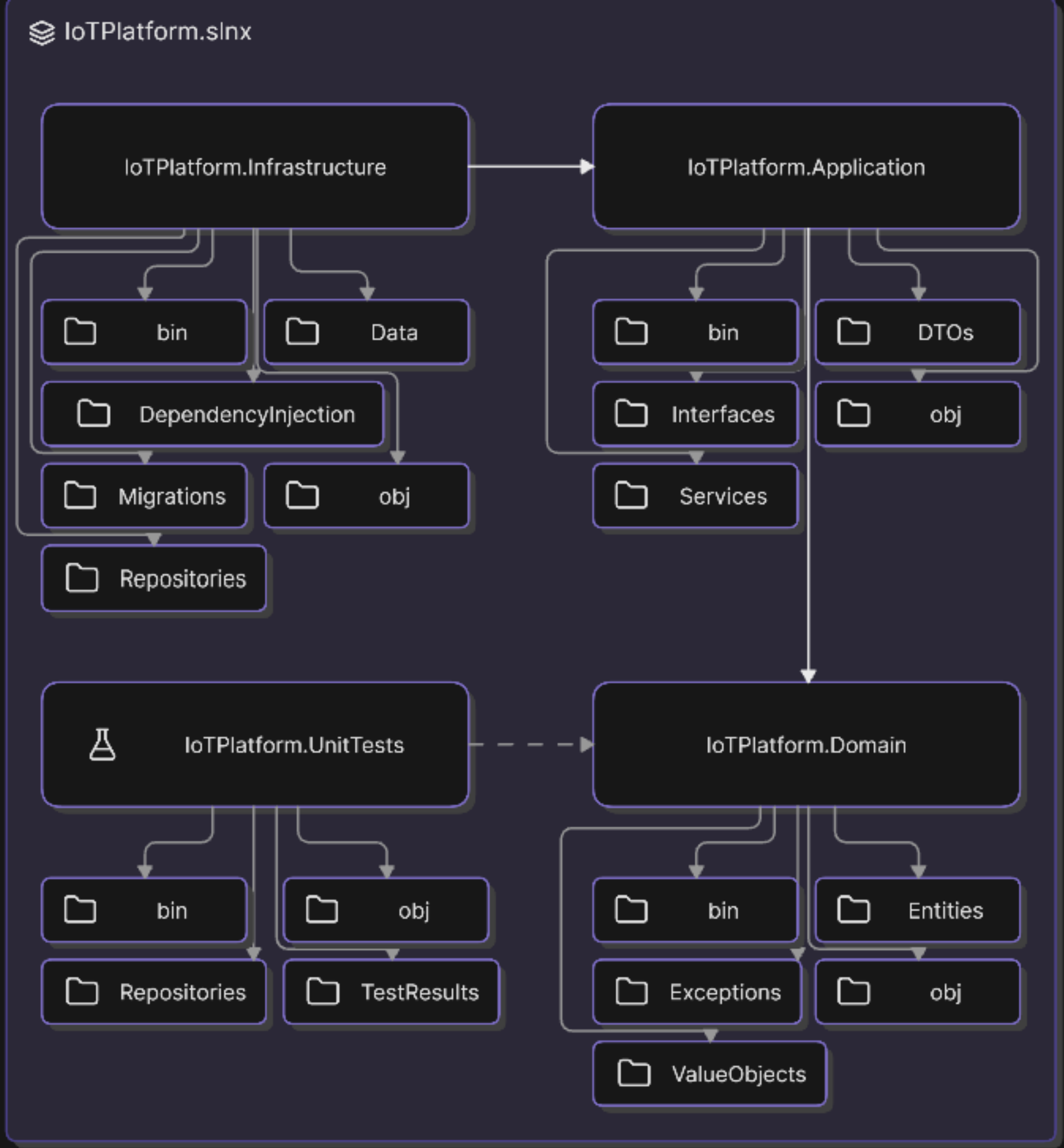
Makayla Baker 8:19 PM

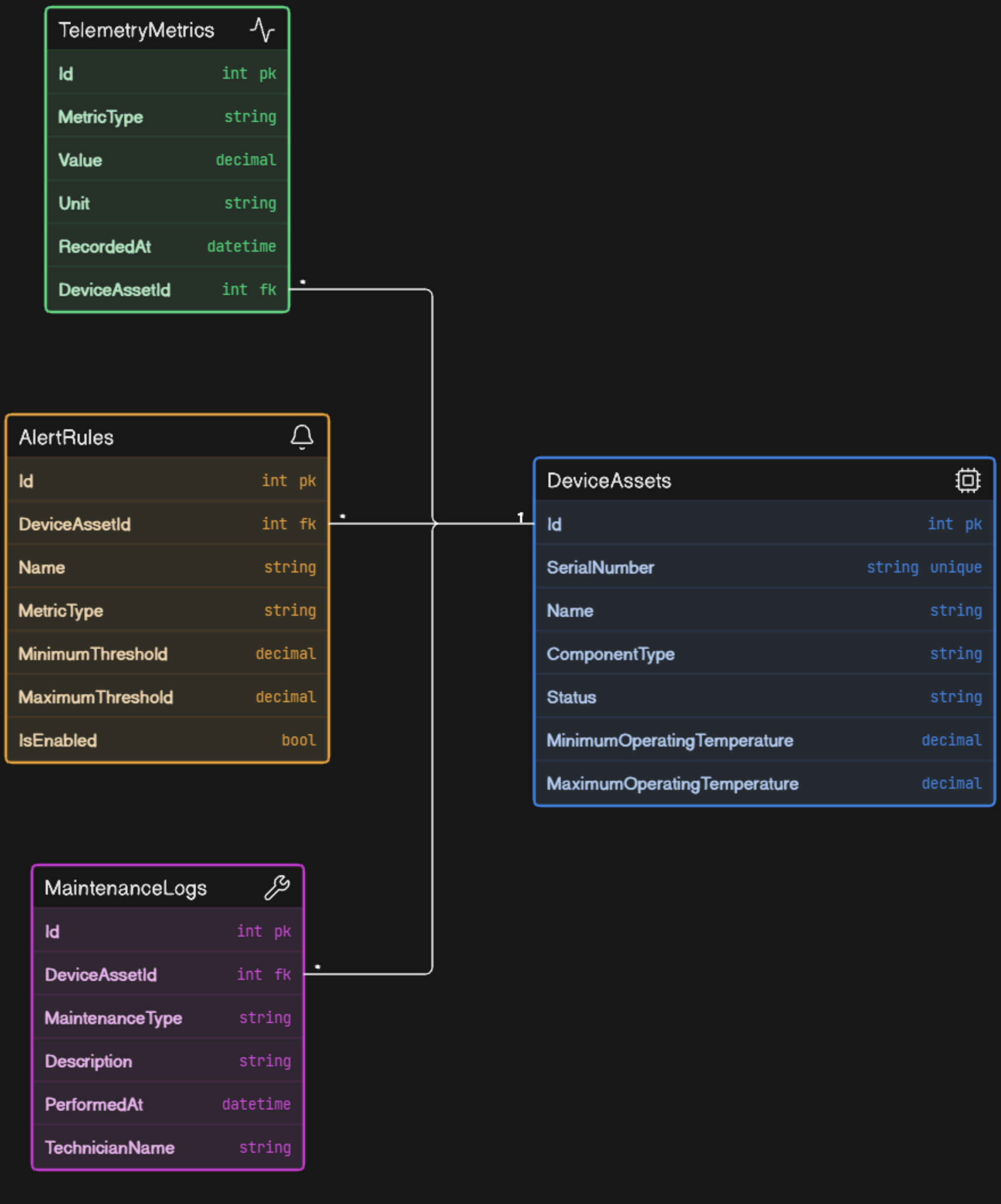
PROBLEM STATEMENT: Aircraft engines and critical components require regular monitoring and maintenance to ensure safe and reliable operation. However, managing component information, operating measurements, and maintenance records separately can make it difficult to identify potential issues and track asset conditions. The Aviation IoT Asset & Telemetry Platform addresses this by providing a centralized system to organize aircraft asset information and monitor recorded telemetry measurements.



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PLATFORM SCOPE: The platform manages aircraft components and their associated telemetry and maintenance records. It supports recording measurements such as temperature, vibration, and oil pressure, evaluating readings against defined alert thresholds, and tracking maintenance activities. Its four main entities are DeviceAsset, TelemetryMetric, AlertRule, and MaintenanceLog. The system also enforces business rules for serial-number formats, positive operating ranges, and valid asset-status transitions. To conclude, it provides a structured foundation for aircraft asset management, telemetry monitoring, and maintenance tracking.


IoTPlatform.slnx
IoTPlatform.Infrastructure
IoTPlatform.Application
bin
Data
DependencyInjection
Migrations
obj
Repositories
bin
DTOs
Interfaces
obj
Services
IoTPlatform.UnitTests
IoTPlatform.Domain
bin
obj
Repositories
TestResults
bin
Entities
Exceptions
obj
ValueObjects


TelemetryMetrics
Id int pk
MetricType string
Value decimal
Unit string
RecordedAt datetime
DeviceAssetId int fk
AlertRules
Id int pk
DeviceAssetId int fk
Name string
MetricType string
MinimumThreshold decimal
MaximumThreshold decimal
IsEnabled bool
DeviceAssets
Id int pk
SerialNumber string unique
Name string
ComponentType string
Status string
MinimumOperatingTemperature decimal
MaximumOperatingTemperature decimal
MaintenanceLogs
Id int pk
DeviceAssetId int fk
MaintenanceType string
Description string
PerformedAt datetime
TechnicianName string
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A	B	C
Solution Topography:	Checkboxes	
IoTPlatform.Domain: Pure business entities, value objects, domain exceptions. Zero dependencies.	☑	Done by: Makayla
IoTPlatform.Application: Interfaces for repositories, DTOs, and service contracts.	☑	Done by: Makayla
IoTPlatform.Infrastructure: EF Core DbContext, entity configurations, SQL Server repositories, and JSON seed loaders.	☑	Done by: Makayla
IoTPlatform.UnitTests: MSTest suite covering domain rules and repository contracts.	☑	Done by: Makayla
Domain & Data Scope:		
Entities: DeviceAsset, TelemetryMetric, AlertRule, and MaintenanceLog.	☑	Done by: Daniel
Enforce business invariants: serial numbers must follow format masks, operating limits must be positive ranges, and status transitions must follow lifecycle rules.	☑	Done by: Daniel
Relational mapping: configure primary keys, foreign keys, indexes, and constraints using EF Core Fluent API.	☑	Done by: Daniel
Verification Suite:		
Unit tests achieving coverage across all domain invariant methods.	☑	Done by: Jake
Integration tests using SQLite in-memory or LocalDB verifying repository CRUD operations.	☑	Done by: Jake
Architecture Blueprint Document (Single PDF):		
System Overview: Problem statement and high-level platform scope.	☑	Done by: Makayla
Domain Model Diagram: Class diagram showing entities, value objects, and relationships.	☑	Done by: Jake
Entity Relationship Diagram (ERD): Relational schema detailing keys, constraints, and data types.	☑	Done by: Daniel
Team Contributions Table: Specific responsibilities and git commits attributed to each team member.	☑	Done by: Jake